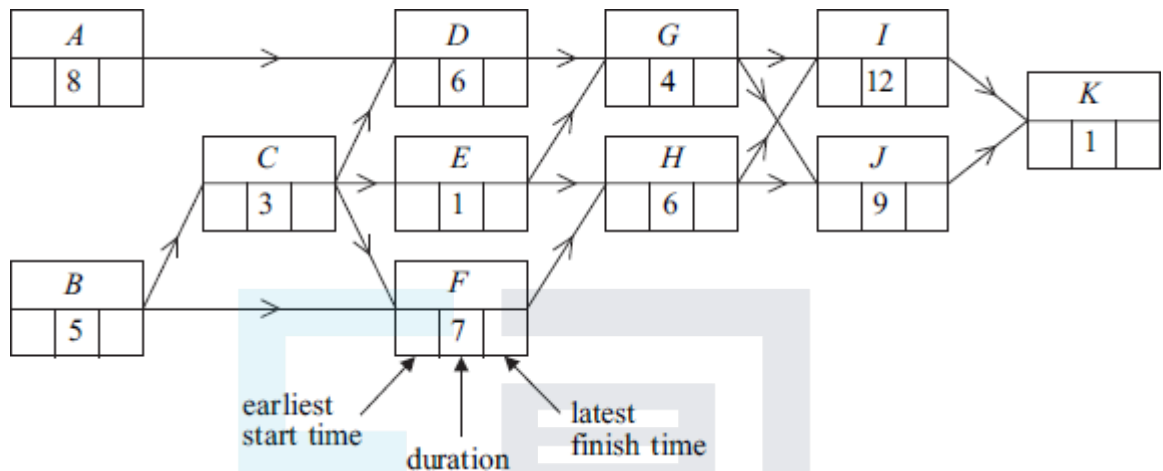


Q1.

Figure 1 below shows an activity diagram for a project. Each activity requires one worker. The duration required for each activity is given in hours.

- (a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**.

Figure 1



(4)

- (b) On **Figure 2** complete the precedence table.

Figure 2

Activity	Immediate predecessor(s)
A	
B	
C	
D	
E	
F	
G	
H	
I	
J	
K	

(2)

- (c) Find the critical path.

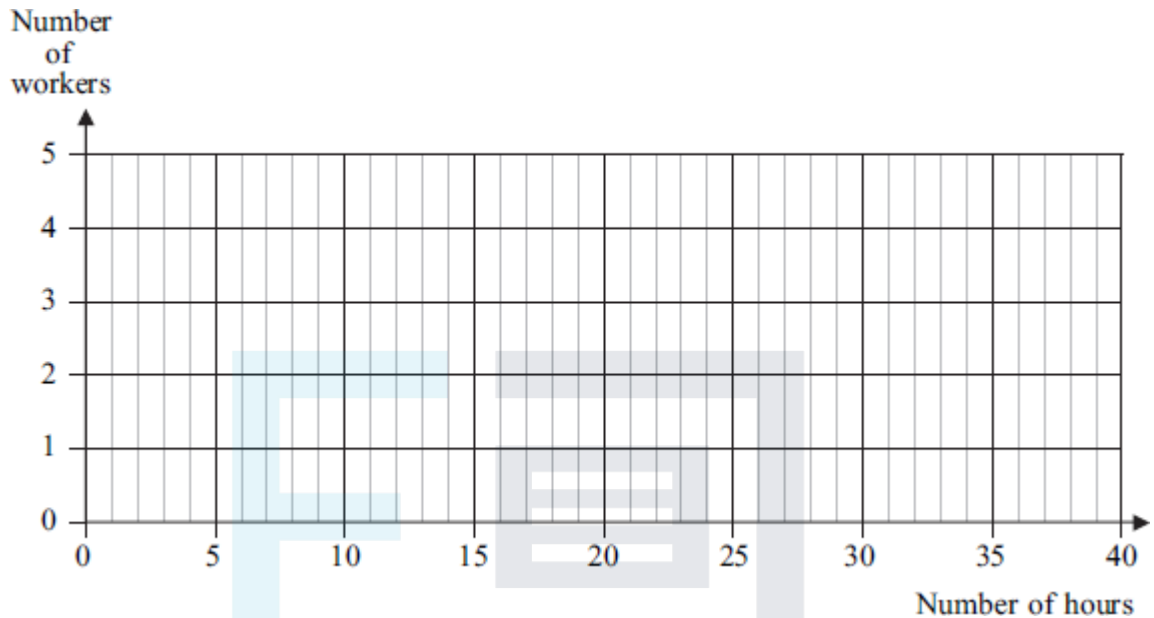
(1)

- (d) Find the float time of activity *E*.

(1)

- (e) Using **Figure 3**, draw a resource histogram to illustrate how the project can be completed in the minimum time, assuming that each activity is to start as early as possible.

Figure 3



(3)

- (f) Given that there are two workers available for the project, find the minimum completion time for the project.

(1)

- (g) Given that there is only one worker available for the project, find the minimum completion time for the project.

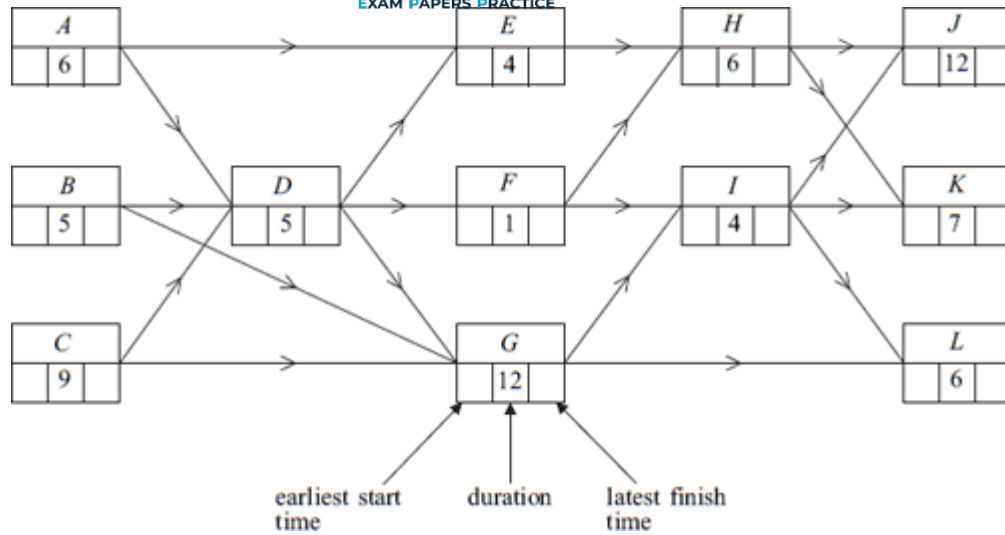
(1)

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(Total 13 marks)

Q2.

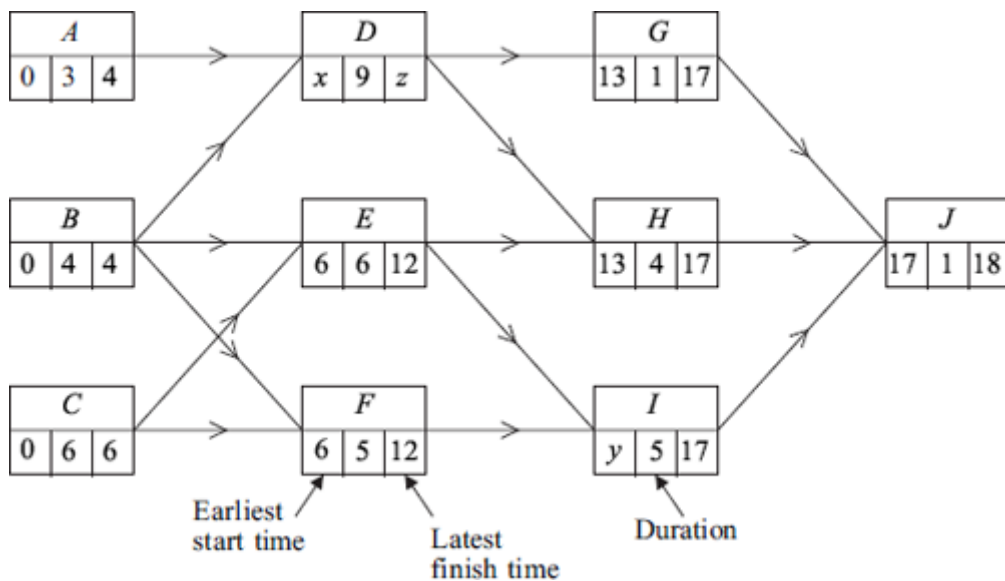
The figure below shows an activity diagram for a project. The duration required for each activity is given in hours. The project is to be completed in the minimum time.



- (a) Find the earliest start time and the latest finish time for each activity and insert their values on the figure. (4)
- (b) Find the critical path. (1)
- (c) Find the float time of activity *E*. (1)
- (d) Given that activities *H* and *K* will both overrun by 10 hours, find the new minimum completion time for the project. (3)
- (Total 9 marks)

Q3.

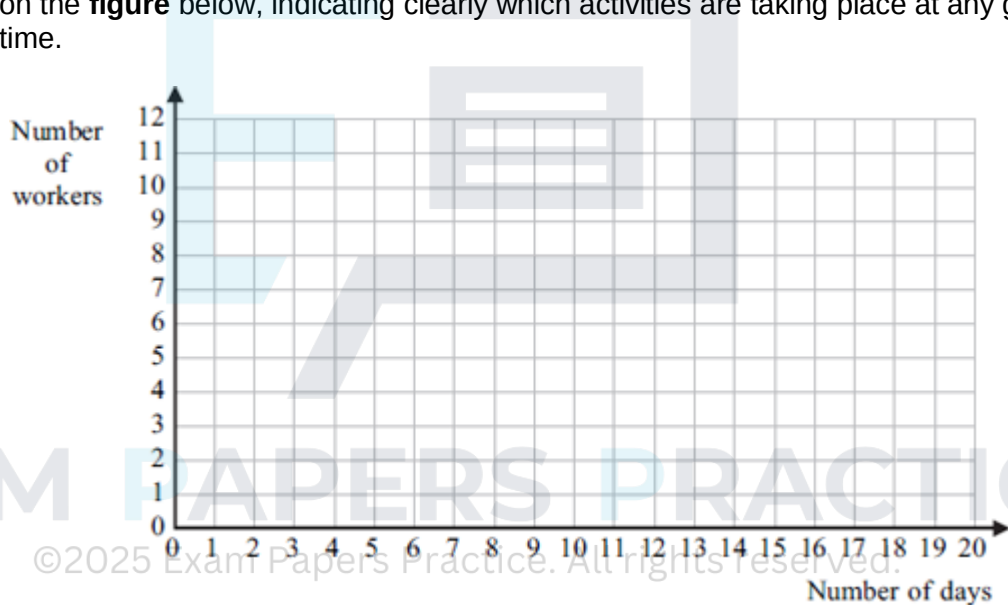
The diagram shows the activity network and the duration, in days, of each activity for a particular project. Some of the earliest start times and latest finish times are shown on the diagram.



- (a) Find the values of the constants x , y and z . (3)
- (b) Find the critical paths. (2)
- (c) Find the activity with the largest float and state the value of this float. (2)
- (d) The number of workers required for each activity is shown in the table.

Activity	A	B	C	D	E	F	G	H	I	J
Number of workers required	4	2	3	4	2	4	3	3	5	6

Given that each activity starts as **early** as possible and assuming that there is no limit to the number of workers available, draw a resource histogram for the project on the **figure** below, indicating clearly which activities are taking place at any given time.



- (e) It is later discovered that there are only 9 workers available at any time. Use resource levelling to find the new earliest start time for activity J so that the project can be completed with the minimum extra time. State the minimum extra time required.

(2)
(Total 14 marks)

Q4.

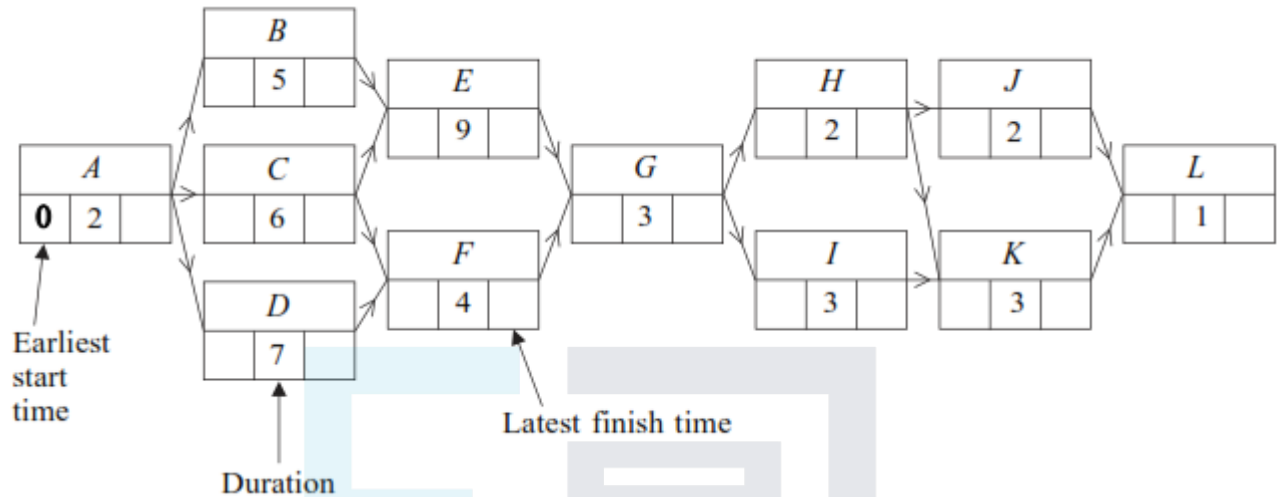
A group of workers is involved in a decorating project. The table shows the activities involved. Each worker can perform any of the given activities.

Activity	A	B	C	D	E	F	G	H	I	J	K	L
Duration (days)	2	5	6	7	9	4	3	2	3	2	3	1

Number of workers required	6	3	5	2	5	2	4	4	5	3	2	4
----------------------------	---	---	---	---	---	---	---	---	---	---	---	---

The activity network for the project is given in **Figure 1** below.

Figure 1



- (a) Find the earliest start time and the latest finish time for each activity, inserting their values on **Figure 1**.

(4)

- (b) Hence find:

- (i) the critical path;
- (ii) the float time for activity *D*.

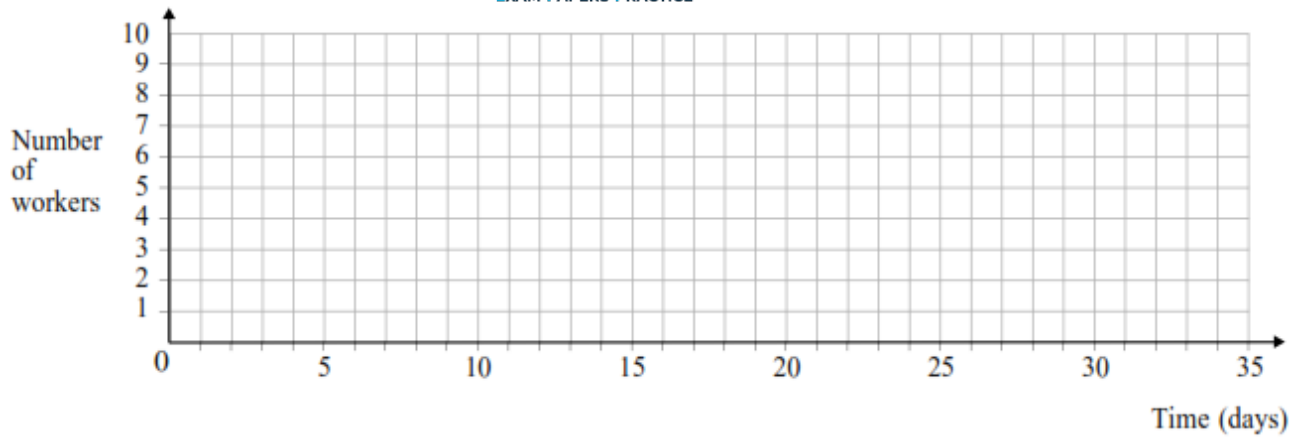
(3)

(i) The critical path is _____

(ii) The float time for activity *D* is _____

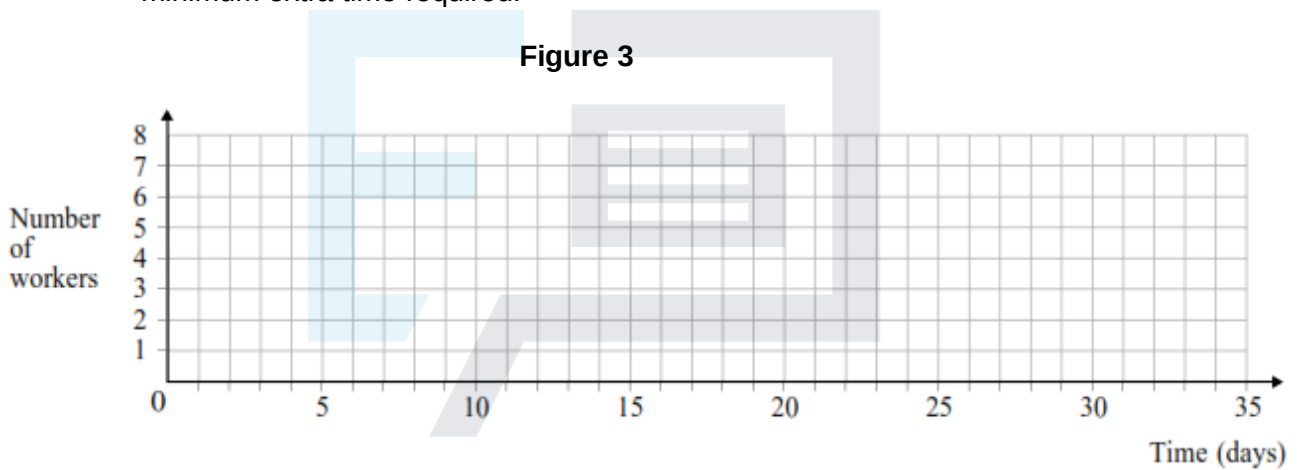
- (c) Given that each activity starts as early as possible and assuming that there is no limit to the number of workers available, draw a resource histogram for the project on **Figure 2** below, indicating clearly which activities are taking place at any given time.

Figure 2



(4)

- (d) It is later discovered that there are only 8 workers available at any time. Use resource levelling to construct a new resource histogram on **Figure 3** below, showing how the project can be completed with the minimum extra time. State the minimum extra time required.



The minimum extra time required is _____

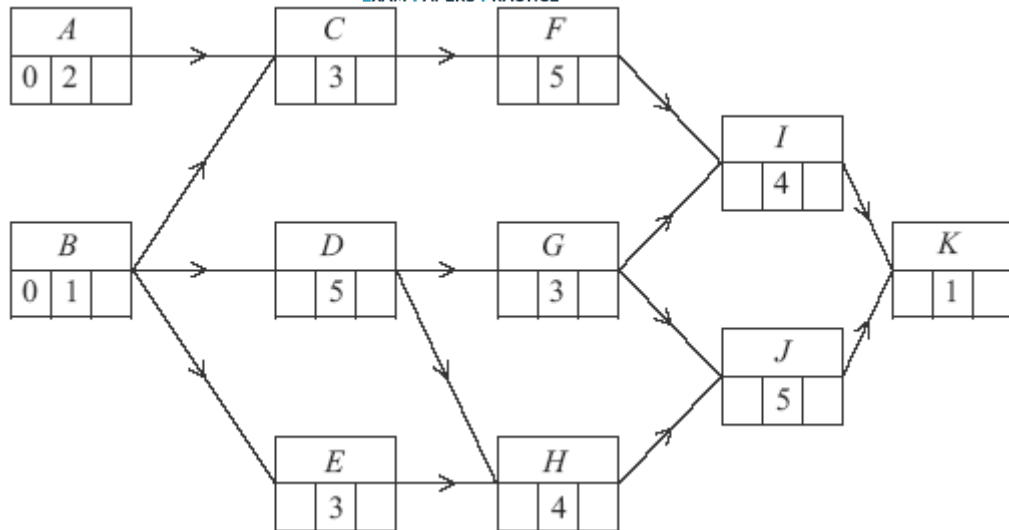
(3)

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(Total 14 marks)

Q5.

The following diagram shows an activity diagram for a building project. The time needed for each activity is given in days.

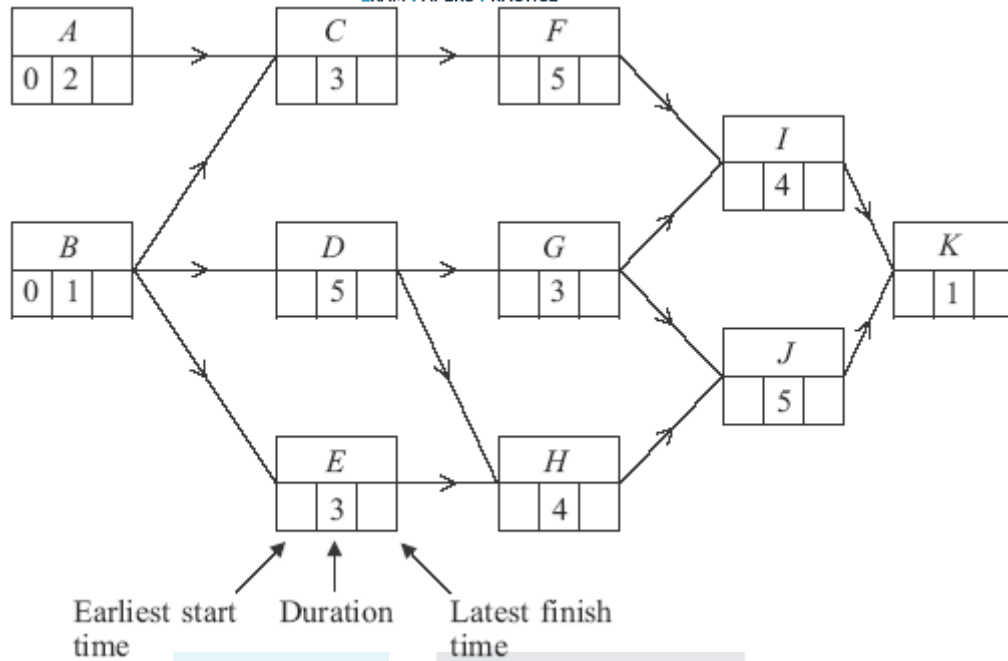


(a) Complete the precedence table below for the project.

Activity	Immediate Predecessors
A	—
B	—
C	
D	
E	
F	
G	
H	
I	
J	
K	

(2)

(b) Find the earliest start times and latest finish times for each activity and insert their values on the diagram below.



(4)

(c) Find the critical path and state the minimum time for completion of the project.

(2)

(d) Find the activity with the greatest float time and state the value of its float time.

(2)

(Total 10 marks)

Q6.

A building project is to be undertaken. The table shows the activities involved.

Activity	Immediate Predecessors	Duration (days)	Number of Workers Required
A	—	2	3
B	A	4	2
C	A	6	1
D	B, C	8	3
E	C	3	2
F	D	2	2
G	D, E	4	2
H	D, E	6	1
I	F, G, H	2	3

- (a) Complete the activity network for the project on **Figure 1**.

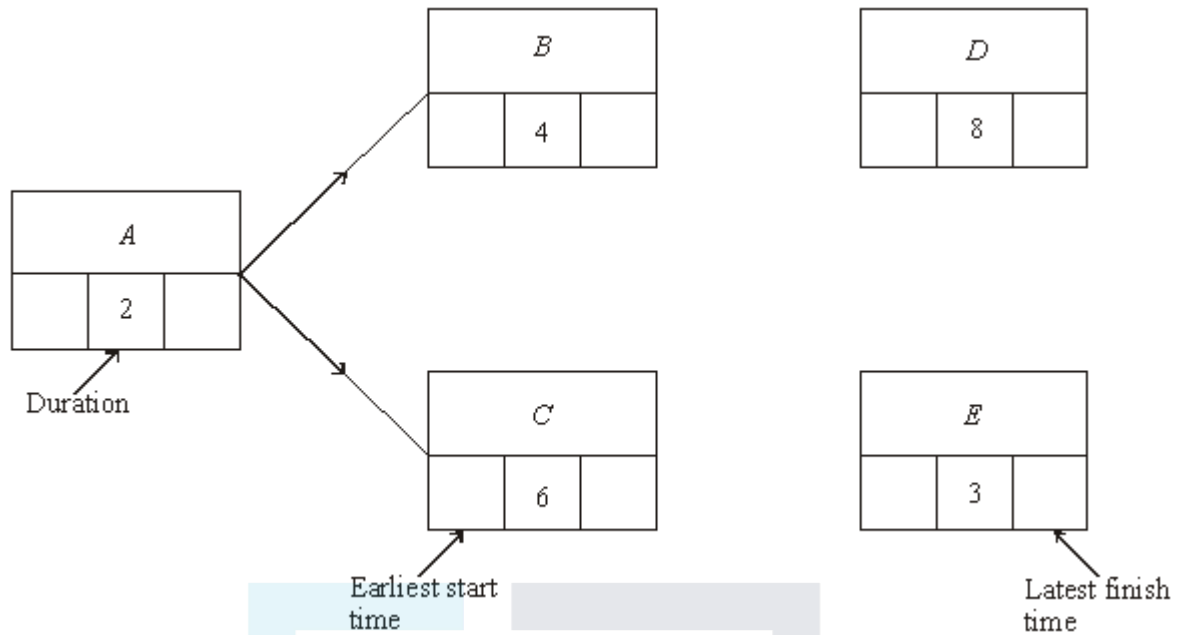


Figure 1

- (b) Find the earliest start time for each activity.

- (c) Find the latest finish time for each activity.

- (d) Find the critical path and state the minimum time for completion.

- (e) State the float time for each non-critical activity.

- (f) Given that each activity starts as early as possible, draw a resource histogram for the project on **Figure 2**.

Number of workers

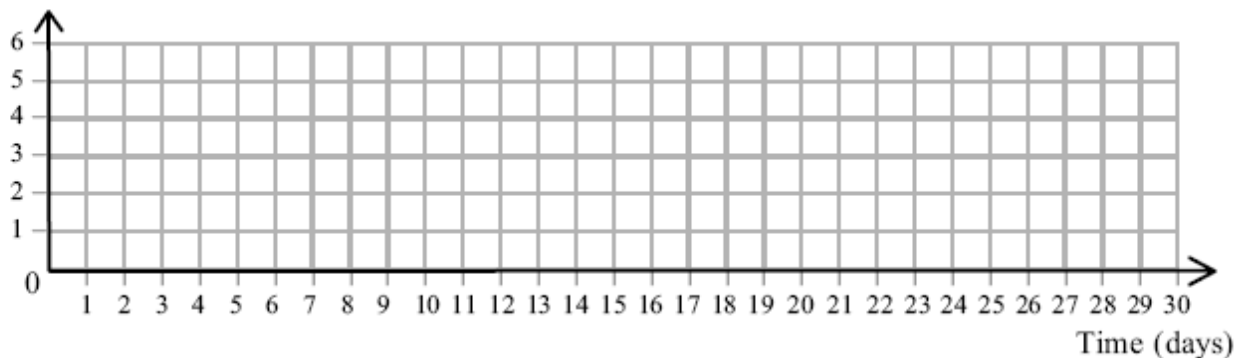


Figure 2

- (g) There are only 3 workers available at any time. Use resource levelling to explain why the project will overran and state the minimum extra time required.

(3)

(Total 18 marks)

Q7.

[Figures 1 and 2, below, are provided for use in this question.]

Figure 1

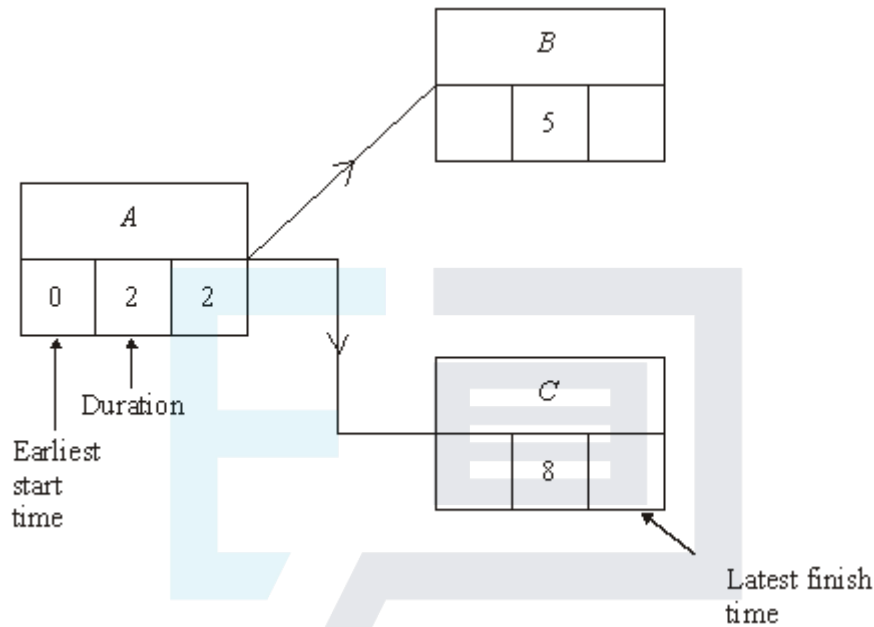
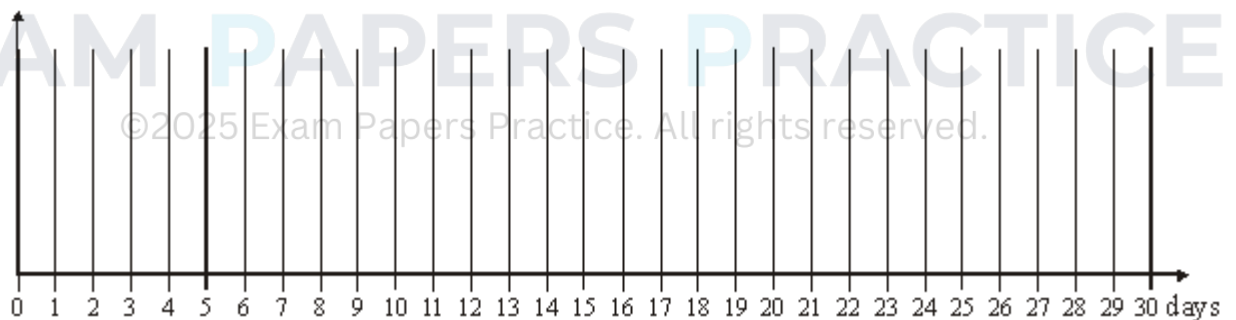


Figure 2



A construction project is to be undertaken. The table shows the activities involved.

Activity	Immediate Predecessors	Duration (days)
A	–	2
B	A	5
C	A	8

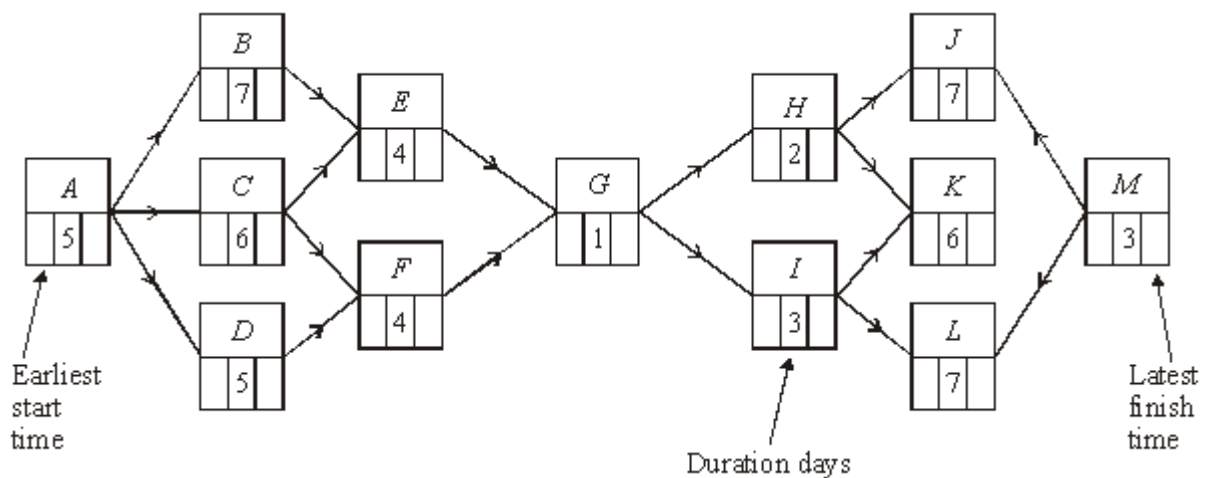
<i>D</i>	<i>B</i>	8
<i>E</i>	<i>B</i>	10
<i>F</i>	<i>B</i>	4
<i>G</i>	<i>C, F</i>	7
<i>H</i>	<i>D, E</i>	4
<i>I</i>	<i>G, H</i>	3

- (a) Complete the activity network for the project on **Figure 1**. (3)
- (b) Find the earliest start time for each activity. (2)
- (c) Find the latest finish time for each activity. (2)
- (d) Find the critical path. (1)
- (e) State the float time for each non-critical activity. (2)
- (f) On **Figure 2**, draw a cascade diagram (Gantt chart) for the project, assuming each activity starts as **late** as possible. (4)
- (Total 14 marks)

Q8.

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The following diagram shows an activity network for a major project.

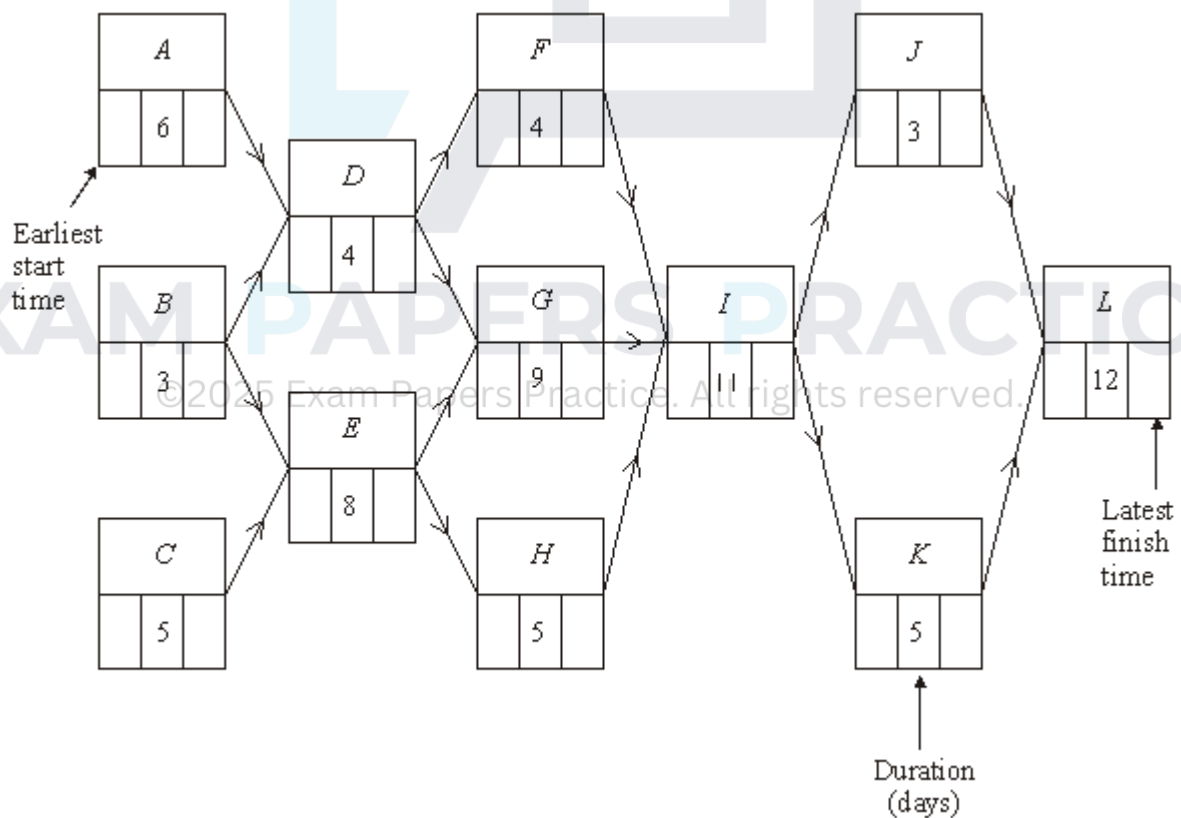


- (a) On the figure above:

- (i) find the earliest start time for each activity; (2)
- (ii) find the latest finish time for each activity. (2)
- (b) Identify the critical path. (1)
- (c) Write down the activity that has a float time of more than 1 day. (1)
- (d) Given that activity *C* overruns by 3 days, find the overrun time for the whole project. (2)
- (e) Given that activities *C* and *F* overrun by 3 days each, find the overrun time for the whole project. (3)
- (Total 11 marks)**

Q9.

The following diagram shows an activity network for a major building project.



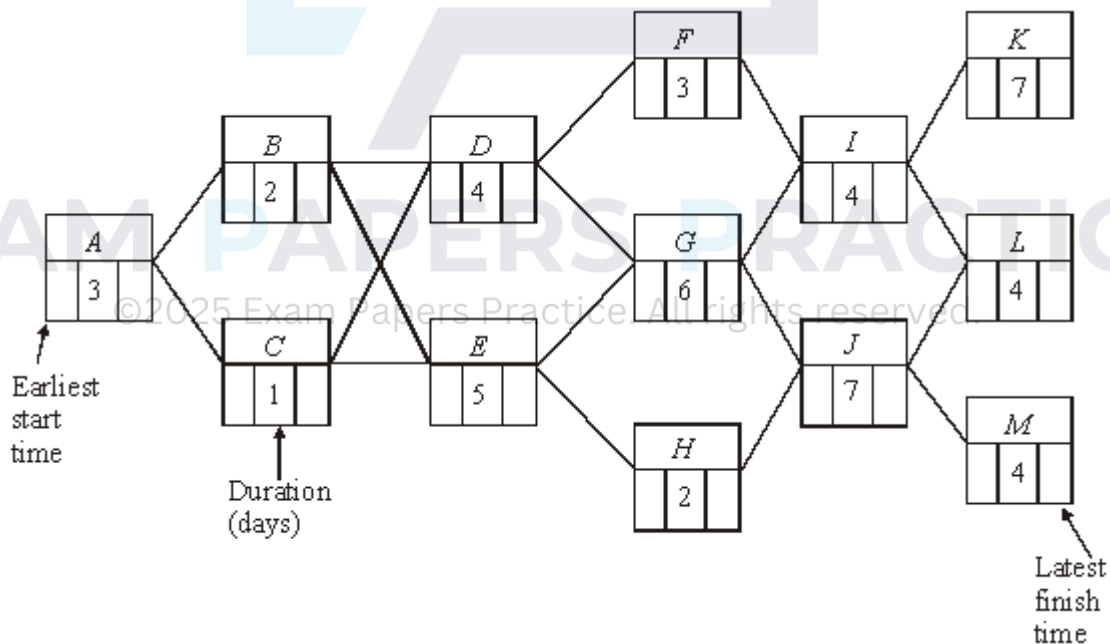
- (a) On the figure above find:
- (i) the earliest start time for each activity; (2)
- (ii) the latest finish time for each activity.

- (b) Identify the critical path. (2)
- (c) State the activity with the greatest float time. (1)
- (d) Construct a Gantt (cascade) diagram for the project. (1)
- (e) Activities *D* and *J* overrun by 5 days each. (3)
- Find:
- (i) the new minimum completion time for the project; (3)
- (ii) the new critical path. (1)

(Total 13 marks)

Q10.

A major construction project is to be undertaken and the project has been divided into 13 activities. The following diagram shows the activity network for the project.



- (a) On the diagram above:
- (i) find the earliest start time for each activity; (2)
- (ii) find the latest finish time for each activity. (2)
- (b) List the non-critical activities.

(1)

- (c) Draw a Gantt (cascade) diagram for the project.

(3)

- (d) There are only two workers available, each able to complete any of the activities in the stated time.

- (i) Find the new minimum completion time for the whole project.

(2)

- (ii) Allocate the activities to the two workers so that this new minimum time can be achieved.

(2)

(Total 12 marks)

Q11.

[Graph paper is provided for use in answering this question.]

A small building project is to be undertaken. The following precedence table shows each activity, its duration, and the number of workers required to complete the activity.

Activity	Immediate predecessor	Duration (days)	Number of workers
<i>A</i>	-	3	4
<i>B</i>	<i>A</i>	4	2
<i>C</i>	<i>A</i>	2	1
<i>D</i>	<i>B</i>	3	2
<i>E</i>	<i>D</i>	11	4
<i>F</i>	<i>D</i>	4	2
<i>G</i>	<i>C, D</i>	5	2
<i>H</i>	<i>F, G</i>	2	1
<i>I</i>	<i>E, H</i>	2	4

- (a) Construct an activity network for the project.

(3)

- (b) Find the earliest start time for each activity.

(2)

- (c) Find the latest finish time for each activity.

(2)

- (d) State the float time for each non-critical activity. (2)
- (e) Given that each activity starts as early as possible, draw a resource histogram for the project. (4)
- (Total 13 marks)**

Q12.

[One sheet of 2 mm graph paper is provided for use in answering this question.]

Each year the secretary of a Mathematical Society undertakes the task of preparing and distributing the society's annual programme. The programme gives dates of meetings, a list of speakers and the titles of their talks, and an up-to-date list of paid-up members. The table below shows the activities which the secretary needs to carry out in preparing and distributing the programme.

	Activity	Immediate predecessor	Duration (days)
<i>A</i>	Select dates for meetings	—	3
<i>B</i>	Book speakers	<i>A</i>	13
<i>C</i>	Obtain advertising material for programme	<i>A</i>	12
<i>D</i>	Send out membership renewal notices	—	21
<i>E</i>	Prepare list of paid-up members	<i>D</i>	6
<i>F</i>	Prepare programme for printers and check proofs	<i>B, C, E</i>	10
<i>G</i>	Assemble material for circulation	<i>F</i>	7
<i>H</i>	Print members' address labels	<i>E</i>	4
<i>I</i>	Send out programmes	<i>G, H</i>	3

- (a) Construct an activity network for the project. (4)
- (b) On your network show full details of the earliest and latest starting and finishing times for each activity. Indicate and state the critical path. (6)
- (c) For each activity state the float time. (2)
- (d) Construct a cascade diagram for the project.
If each activity requires one helper so that activities are completed on time, state the minimum number of helpers required for the project. (5)
- (e) The secretary is let down by one helper who resigns immediately prior to the start of the project and so cannot be replaced. Find the least possible number of days' delay to the completion of the project and explain how it can be achieved. (2)

(Total 19 marks)